

Kassidy Barram, Software Engineer

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LINKS

[LinkedIn](#), [Personal GitHub](#), [Personal Website](#)

PROFESSIONAL SUMMARY

Software engineer with 4.5+ years of experience designing and building distributed systems and decentralized compute infrastructure. Expertise setting up, managing, and building in both public and private cloud environments. Extensive exposure to agile software engineering practices, building web applications within big data spaces, and cross-functional team dynamics and collaboration. Master's degree in computer science with multiple publications and research awards.

EMPLOYMENT HISTORY

Sept 2025 — Present	Software Engineer, qBitTensor Labs	Boulder, CO
	• Architected a fault-tolerant, distributed validation platform in Python across over 250 independent nodes. • Led development of public and private miner codebases, providing the foundation for participants to submit quantum-inspired solutions to NP-hard optimization problems. • Engineered a security-in-depth Docker-based sandboxed execution environment for untrusted third-party code. • Built and managed data pipelines and persistence layers using MySQL, SQLite, and SQLAlchemy to support solution tracking, scoring, and network state management.	
Oct 2021 — Mar 2025	Software Engineer, Project Sustain	
	• Designed and built 4 spatio-temporal data visualization web applications in React and deck.gl backed by PostGIS and MongoDB, facilitating high-throughput, low-latency geospatial data rendering across distributed environments. • Architected a resilient replacement for brittle, failure-prone infrastructure as Kubernetes administrator, enabling continuous deployment, scaling, and scheduling of 24+ distributed services at 99.9% uptime. • Administered 200 node private cloud. • Built 2 distributed systems for ML model validation, supporting linear regression and classification across a 75-machine compute network.	
May 2019 — Aug 2019	Systems Engineer Intern , CPP Wind Engineering Consultants	
	• Conducted quantitative testing to evaluate accuracy and frequency response across multiple sensor data collection methods, informing instrument selection and measurement reliability. • Designed a control station that mixed and regulated the output of tracer gas used in the wind tunnel testing of the air quality from various exhausts produced by buildings. • Built and assembled city-scape proximity models for data collection in wind tunnels	

EDUCATION

Aug 2022 — May 2025	Master of Science, Colorado State University	Fort Collins
	• Graduated <i>Summa cum Laude</i> (4.0 GPA) • Awarded Best Paper at 2023 International ACM/IEEE Big Data Conference. • Published Research in 4 competitive (IEEE/ACM) international computer science conferences. • Won award at CURC (Celebrating Undergraduate Research and Creativity).	
Aug 2017 — May 2022	Bachelor of Science, Colorado State University	Fort Collins
	• Graduated <i>Cum Laude</i> (3.95GPA) • Awarded National Science Foundation Research Grant to research distributed systems and big data technologies. • Participated in research to facilitate communication between Pixhawk 4 Flight Controller and the Inertial Measurement Unit attached to drones using C++.	

SKILLS

Java	GitHub
Python	Big Data
C++ / C	Distributed Systems

JavaScript / TypeScript

Docker

Kubernetes

MongoDB

PyTorch / PySpark

gRPC

Hadoop MapReduce

Apache Spark

Tensor Flow

TCP Networking

Test Driven Development

Machine Learning

Artificial Intelligence

RESTful API

Agile

CI/CD Pipelines

Prometheus

Grafana

PUBLICATIONS

- Archimedes: A Framework to Support Distributional Similarity Analysis over Arbitrary Spatiotemporal Scopes at Scale, *IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2025. Paige Hansen, Nathan Orwick, **Kassidy Barram**, Pierce Smith, Jay Breidt, Sangmi Lee Pallickara, Shrideep Pallickara
- Scribe: Enabling Programmatic Interfaces for Explorations Over Voluminous Spatiotemporal Data Collections, *IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT)*, 2024. **Kassidy Barram**, Sangmi Lee Pallickara, Shrideep Pallickara
- A Framework for Profiling Spatial Variability in the Performance of Classification Models, *IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT)*, 2023. **Best Paper Award**. Menuka Warushavithana, **Kassidy Barram**, Saptashwa Mitra, Caleb Carlson, Sudipto Ghosh, Jay Breidt, Sangmi Lee Pallickara, Shrideep Pallickara
- Resource Efficient Profiling of Spatial Variability in Performance of Regression Models, *IEEE International Conference on Big Data (IEEE BigData)*, 2022. Caleb Carlson, Menuka Warushavithana, Saptashwa Mitra, **Kassidy Barram**, Sudipto Ghosh, Jay Breidt, Sangmi Lee Pallickara, Shrideep Pallickara